



AI inference on Android

Android Lecture 6

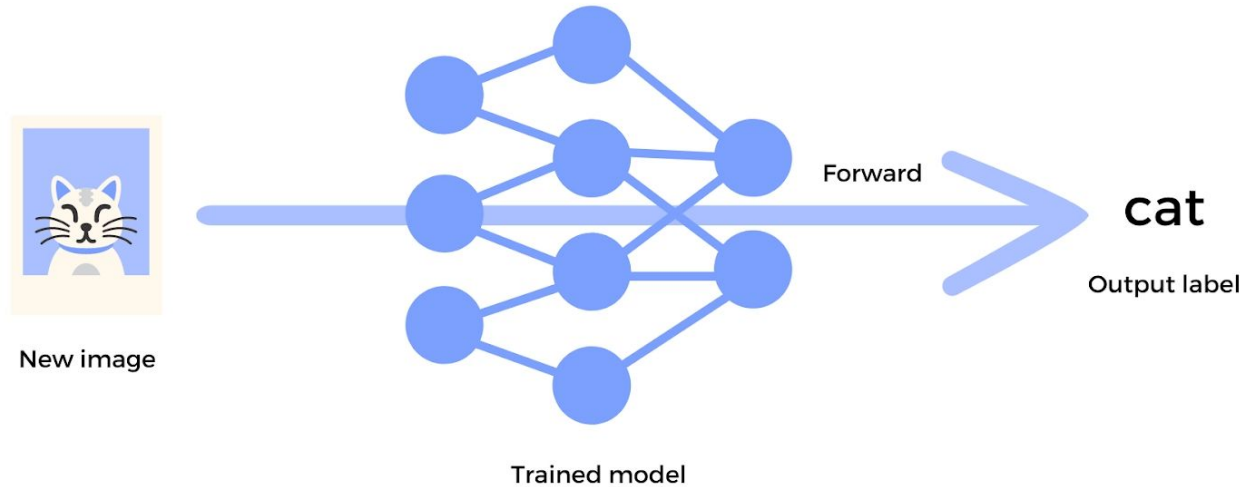
AI model

- *Black box*
- *Function: (Input) -> Output*
- *Most common implementation in modern AI: Deep Neural networks (DNN)*



What is inference

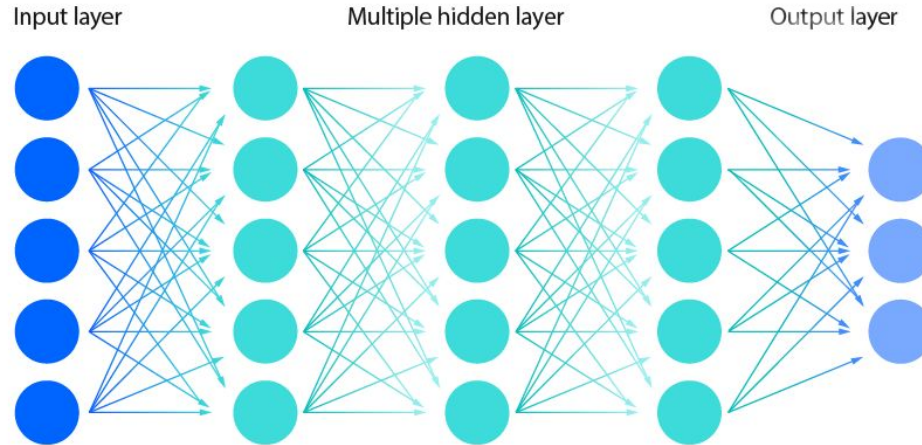
Inference is the process of using a trained machine learning model to make predictions on new data.



Deep Neural Network

Network of interconnected neurons (smallest computational unit) organized into multiple layers.

Training: *An optimization problem; finding the parameters of a function that minimizes a loss.*



Adding model to an Android application

01

Choose AI Model

- Use existing pre-trained model, or
- Fine-tune existing model, or
- Train model from scratch

02

Integrate model to app

- **Convert** to compatible format if needed
- Choose **distribution**: encapsulate within .apk, remote distribution

03

Run model on-device

- Choose correct **Runtime environment**
- **Load model** and **run inference** when needed

AI technologies for Android (on-device)

Runtime environments:

- Tensorflow Lite
- LiteRT
- MediaPipe LLM Inference API
- ML Kit
- ML Kit GenAI
- NNAPI (Android Neural Networks API)
- ONNX Runtime Mobile (Cross platform)
- Google AI Edge SDK

Pre-trained models:

- Gemini Nano
- Gemma models
- Community available models
 - <https://huggingface.co/litert-community>
 - <https://www.kaggle.com/>

Model training:

- TensorFlow framework
- PyTorch
- JAX

AI technologies for Android (on-device)

Runtime environments:

- ~~Tensorflow Lite~~ **DEPRECATED**
- LiteRT - replaces Tensorflow Lite
- MediaPipe LLM Inference API
- ML Kit
- ML Kit GenAI **BETA**
- ~~NNAPI (Android Neural Networks API)~~ **DEPRECATED**
- ONNX Runtime Mobile (Cross platform)
- Google AI Edge SDK **EXPERIMENTAL**

Pre-trained models:

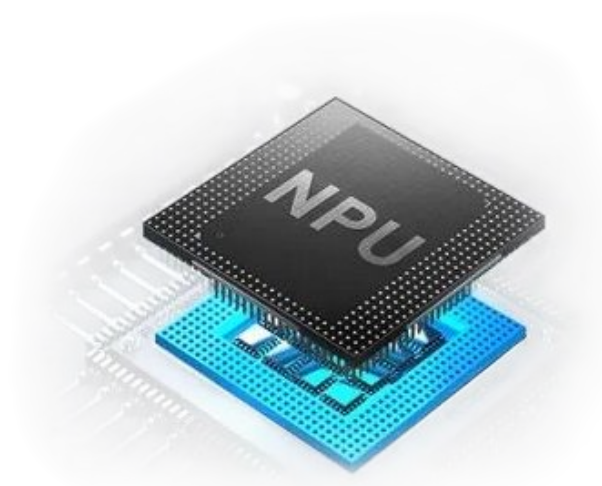
- Gemini Nano **EXPERIMENTAL**
- Gemma models
- Community available models
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 - <https://www.kaggle.com/>

Model training:

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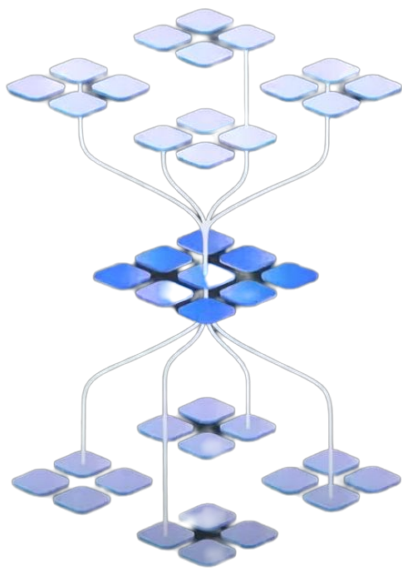
NPU

- Neural processing unit
- Training and execution of AI models
- Optimized for matrix multiplication
- Massive parallelization
- Contains on-chip high-bandwidth memory (SRAM)
- More efficient for AI than CPUs: lower battery usage, faster computation, more parallelism



Gemini Nano

- Lightweight version of Gemini that can run on-device and offline
- Delivered through the Android AICore runtime
- Limited device support:
 - Android 14+
 - Pixel 8+, some Galaxy phones
- Supports *Low Rank Adaptation* (LoRA)
- Used by Google in Smart replies
- Runtime:
 - [ML Kit GenAI APIs](#) (beta)
 - [Google AI Edge SDK](#) (experimental)





- A collection of lightweight, state-of-the-art open models
- *Robust models available for using as is or with fine-tuning*
- Developed by Google DeepMind
- *Runtime environments:*
 - [MediaPipe LLM Inference API](#) (available for Android and iOS)
 - After conversion also [TFLite](#)
- [Models:](#)
 - Gemma 3
 - CodeGemma
 - PaliGemma 2
 - ShieldGemma 2



- *Provides runtime for model inference*
- *Previously TensorFlow Lite (TFLite)*
- *Uses .tflite format of models*
- Model can be distributed using Google services ([source](#))
- Support for on-device training ([source](#))
- [LiteRT Next](#): new API in Alpha
 - Takes advantage of NPU ([source](#))
- Support:
 - API 21+ (Android 5) for Android
 - iOS
 - Embedded Linux
 - Microcontrollers
- Example of usage: [Audio classification](#)



Initialisation and inference

```
private lateinit var interpreter: InterpreterApi

val initializeTask: Task<Void> = TfLite.initialize(this)

initializeTask.addOnSuccessListener {
    val interpreterOption = InterpreterApi.Options().setRuntime(TfLiteRuntime.FROM_SYSTEM_ONLY)
    interpreter = InterpreterApi.create(
        modelBuffer,
        interpreterOption
    )
    .addOnFailureListener { e ->
        Log.e("Interpreter", "Cannot initialize interpreter", e)
    }
}

initializeTask.await()

interpreter.run(inputBuffer, outputBuffer) // Run model
```



- *One of the most known and used Machine learning framework*
- *Mostly used in Python*
- *Standard Tensorflow model can be converted to .tflite and run using LiteRT ([source](#))*
- *Allows creating custom NNs using predefined or custom layers of neurons*
- *Fine-tuning existing models, for example:*
 - **BERT** (Bidirectional Encoder Representations from Transformers) for natural language processing
 - **ResNet** (Residual network) for image classification
 - **Wav2Vec2** for speech to text
 - **GPT** models for text-based tasks
 - ...

How to train neural network on data set

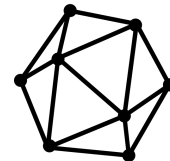
1. *Define problem: input/output*
2. *Create or obtain data set (and divide to - train, validation and test sets)*
3. *Choose technology/framework*
4. *Design model: number of layers and neurons, types of layers, interconnections, activation functions, etc.*
5. *Train using data set:*
 - a. *Train set used for learning*
 - b. *Validation for testing during learning*
 - c. *Test data evaluate performance at the end of training*
6. *(Optional) Optimise - pruning, reduce bits used for parameters, ...*
7. *Export model*
8. *(Optional) Convert to needed format*

Advantages:

- *Privacy & data security*
- *Low cost*
- *Internet connection not needed*
- *Energy efficient*
- *Can be quick for small models on NPU*

Disadvantages:

- *Model size*
- *Old/slow devices (performance)*
- *Updating of model is hard*
- *No central learning for model*
- *NPU for optimisation is needed (often required)*
- *Compatibility issues*
- *Limited resources for robust models*
- *Battery consumption*
- *Relatively new technology with small community support*



Links & Sources:

- Hugging Face LiteRT community <https://huggingface.co/litert-community>
- Kaggle community (articles, sharing, models): <https://www.kaggle.com/>
- Gemini Nano on Android Developer — <https://developer.android.com/ai/gemini-nano>
- Gemini Nano (Markovate blog) — <https://markovate.com/blog/gemini-nano/>
- A New Foundation for AI on Android (Android Developers Blog) — https://android-developers.googleblog.com/2023/12/a-new-foundation-for-ai-on-android.html?utm_source=chatgpt.com
- Low Rank Adaptation: [datacamp LRA](#)
- Gemma Mobile — <https://ai.google.dev/gemma/docs/integrations/mobile>
- Gemma models — <https://ai.google.dev/gemma/docs>
- CPU, GPU, and NPU: Understanding Key Differences and Their Roles in AI — <https://levysoft.medium.com/cpu-gpu-and-npu-understanding-key-differences-and-their-roles-in-artificial-intelligence-2913a24d0747>
- TensorFlow Lite (TensorFlow GitHub) — <https://github.com/tensorflow/tensorflow/tree/master/tensorflow/lite>
- LiteRT (Google AI Edge) — <https://ai.google.dev/edge/litert>
- LiteRT Android Kotlin (Google AI Edge) — https://ai.google.dev/edge/litert/next/android_kotlin
- Tensorflow — <https://www.tensorflow.org/>
- Tensorflow tutorials — <https://www.tensorflow.org/tutorials>
- ONNX runtime — <https://onnxruntime.ai/docs/tutorials/mobile/>
- Google machine learning courses — <https://developers.google.com/learn?category=aiandmachinelearning>

Thank you for attention

Feel free to leave feedback about this lesson:

